

When AI Exposure Rankings Disagree: Robust Top-25 Selection Across 661 U.S. Occupations

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Abstract—AI-exposure measures do not necessarily identify the same occupations as priorities. This creates a practical problem when institutions must choose a fixed set of occupations for workforce screening and further occupational assessment. Using 661 U.S. occupations and 18 coherent AI-exposure specifications, we ask which occupations remain stable priorities and which fixed Top-25 list is least sensitive to the admitted measurement-design family. Under coherent designs, 14 occupations appear in every Top-25 and 37 appear in at least one; the more conservative independent interval box gives only 4 guaranteed and 87 possible members. A relative-minimax list raises worst-scenario capture from 93.70% for the baseline composite to 96.52%, although its exact independent-box absolute regret is slightly larger (0.01945 versus 0.01810). We derive exact best/worst rank certificates, a lossless existence-based screening theorem, and an exact fixed-list box-regret corner formula. A nested-support sensitivity then shows where membership conclusions change as the box contracts toward a declared point core. The reported values $g^* = 0.574320, 0.434651$, and 0.078756 are transition boundaries: because best rank uses a strict comparison and worst rank a non-strict comparison, post-contraction membership applies immediately below the relevant boundary rather than being asserted at equality. The resulting Top-25 is a screening and further-assessment list, not a displacement forecast, transition probability, or program-success prediction.

Index Terms—AI exposure, occupational prioritization, robust Top- k , uncertain ranking, minimax regret, workforce screening

I. RESULTS FIRST: WHAT CHANGES FOR WORKFORCE SCREENING?

Occupational AI-exposure rankings can vary materially with measurement design, so a single printed ranking can conceal decision instability [1]. At $k = 25$, only 14 of 661 occupations occur in every coherent Top-25 list, while 37 occur in at least one. The independent coordinatewise interval box is intentionally more conservative: it certifies only 4 guaranteed members and 87 possible members. Table I converts those counts into a direct screening interpretation.

TABLE I
OCCUPATIONAL STABILITY UNDER COHERENT AI-EXPOSURE
DISAGREEMENT AT $k = 25$.

Status	Interpretation	n
Stable	In every coherent Top-25	14
Design-sensitive	In some but not all coherent Top-25 lists	23
Outside family	In no coherent Top-25 list	624

The stable set spans managerial, professional, service, sales, clerical, transportation, and logistics occupations. Examples include General and Operations Managers, Accountants and Auditors, Registered Nurses, Cashiers, Retail Salespersons, Office Clerks, Heavy and Tractor-Trailer Truck Drivers, and Stockers and Order Fillers; the full 14-occupation set with SOC codes is reported in the Supplement.

When one fixed list is required, the relative-minimax solution retains at least 96.52% of each scenario’s own optimal Top-25 priority mass. The baseline composite retains at least 93.70%. Thus the applied question is not “which ranking is true?” but “which priorities are stable across admitted designs, and what fixed list limits worst-case loss when designs disagree?” The mathematics below supplies exact certificates for that decision.

II. DECISION PROBLEM AND COHERENT UNCERTAINTY

Let $i \in \{1, \dots, n\}$ index occupations. The application uses AI Occupational Exposure (AIOE) as one exposure channel [2] and two Microsoft applicability channels (user-goal and AI-action/nonphysical) [3]. Scale s_i is May 2025 OEWS employment share [4], and the five occupation-level skill descriptors are from O*NET 30.3 [5].

To make the modifier construction explicit, let $a_i^{(r)} \in [0, 1]$ be the frozen normalized exposure value for channel r , and let $z_{im} \in [0, 1]$ be the frozen normalized O*NET Importance value for skill m in occupation i . The five skills are Critical Thinking, Active Learning, Learning Strategies,

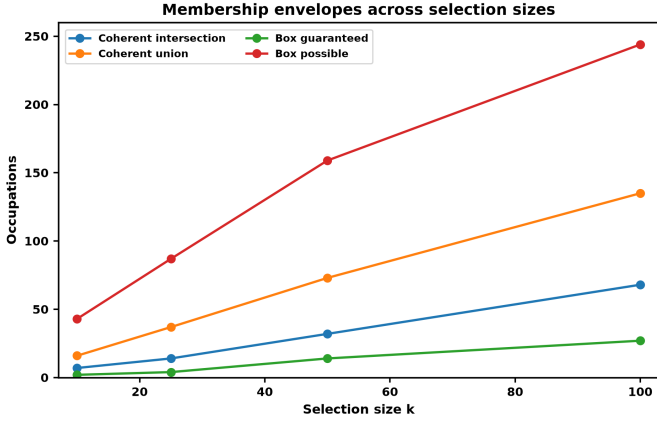


Fig. 1. Membership envelopes across evaluated selection sizes. Coherent specifications move occupational scores together, whereas the independent box permits coordinatewise adversarial movement.

Active Listening, and Speaking. For construction w with retained skill set M_w , define

$$h_i^{(w)} = \frac{1}{|M_w|} \sum_{m \in M_w} z_{im}, \quad (1)$$

where the equal-five construction uses all five skills and each of the five ablations removes exactly one skill. The exposure-conditioned modifier and employment-scaled priority are

$$q_i^{(r,w)} = a_i^{(r)} (1 - h_i^{(w)}), \quad p_i^t = s_i q_i^{(r,w)}, \quad (2)$$

with $t = (r, w)$. The factor $1 - h_i^{(w)}$ is a declared screening transformation: a larger retained-skill aggregate lowers the modifier within this construction, but it is not causal evidence that skills protect a worker from AI exposure or displacement. The paper treats the normalized exposure and skill inputs as frozen primitive inputs to the decision layer; it does not re-estimate their upstream measurement models. This construction follows transparent composite-indicator and multiverse practice: admitted designs are kept coherent rather than replaced by arbitrary coordinatewise combinations [6], [7], [8], [9], [10].

The application fixes $n = 661$ U.S. occupations and $T = 18$ primitive scenarios: three exposure channels times six skill constructions. For each occupation,

$$\ell_i = \min_t p_i^t, \quad u_i = \max_t p_i^t, \quad (3)$$

and

$$\mathcal{P}_\square = \prod_{i=1}^n [\ell_i, u_i], \quad \mathcal{P}_T = \{p^t : t = 1, \dots, T\} \subseteq \mathcal{P}_\square. \quad (4)$$

The declared point core $c_i = p_i^{\text{AIOE, equal-five}}$ is one primitive scenario, hence

$$\ell_i \leq c_i \leq u_i \quad \forall i. \quad (5)$$

No probability distribution is assigned to either uncertainty set.

III. EXACT INDEPENDENT-BOX RANK CERTIFICATES

Uncertain-ranking research distinguishes items that can attain a rank from items that must remain inside a rank range [11], [12], [13]. Here tie treatment is asymmetric by construction. Best rank is existential and resolves endpoint ties in occupation i 's favor; worst rank is pessimistic and counts ties against i :

$$r_i^{\min} = 1 + \sum_{j \neq i} \mathbf{1}\{\ell_j > u_i\}, \quad (6)$$

$$r_i^{\max} = 1 + \sum_{j \neq i} \mathbf{1}\{u_j \geq \ell_i\}. \quad (7)$$

Here $\mathbf{1}\{\cdot\}$ denotes the indicator function. The endpoint realizations are feasible in $\mathcal{P}_\square$, so the certificates are exact:

$$i \text{ guaranteed Top-}k \iff r_i^{\max} \leq k, \quad (8)$$

$$i \text{ possible Top-}k \iff r_i^{\min} \leq k. \quad (9)$$

Figure 1 shows how the box and coherent envelopes differ over $k \in \{10, 25, 50, 100\}$.

IV. COHERENT FIXED-LIST RELATIVE MINIMAX

The fixed-list decision is a finite minimax-regret selection problem over the declared coherent scenarios [14], [15], [16], [17], [18], [19]. Let $V_k(p)$ be the sum of the k largest components of p , and define $V_t^* = V_k(p^t)$. For all admitted scenarios, $V_t^* > 0$. For binary selection $x_i \in \{0, 1\}$ with $\sum_i x_i = k$, coherent capture and relative regret are

$$C_t(x) = \frac{\sum_i p_i^t x_i}{V_t^*}, \quad R_t(x) = 1 - C_t(x). \quad (10)$$

The fixed-list problem is

$$\min_{x, \rho} \quad \rho \quad (11)$$

$$\text{s.t.} \quad \sum_i p_i^t x_i \geq (1 - \rho) V_t^*, \quad \forall t, \quad (12)$$

$$\sum_i x_i = k, \quad x_i \in \{0, 1\}, \quad 0 \leq \rho \leq 1. \quad (13)$$

Relative-regret and uncertain-return selection formulations provide the closest optimization context [20], [21], [22], [23], [24]. Here ρ is a worst relative shortfall, not a probability.

A. Lossless screening

Let $G = \{i : r_i^{\max} \leq k\}$ and $Z = \{i : r_i^{\min} > k\}$.

Theorem 1 (existence-based lossless screening). Assume $\mathcal{P}_T \subseteq \mathcal{P}_\square$. For any minimization objective $F(M_1(x), \dots, M_T(x))$ that is coordinatewise nonincreasing in scenario-wise selected masses $M_t(x) = \sum_i p_i^t x_i$, there exists an optimum containing every item in G and no item in Z .

Proof sketch. Order the items of G by nonincreasing lower bound and insert them in that order. If the next $a \in G$ is absent from a size- k optimum, then at most $k - 1$ competitors satisfy $u_j \geq \ell_a$, so the list contains some h with $u_h < \ell_a$. Any guaranteed item a' inserted earlier satisfies $\ell_{a'} \geq \ell_a$ and therefore $u_{a'} \geq \ell_{a'} \geq \ell_a$, so it cannot be this

h . Replacing h by a thus preserves all earlier insertions and weakly increases every scenario-wise selected mass. After all G items are inserted, if a selected $i \in Z$ remains, at least k competitors satisfy $\ell_j > u_i$, so some omitted j strictly dominates i in every scenario. Replacing i by j strictly increases every scenario-wise selected mass. Because the family of size- k lists is finite, strict exchanges cannot cycle. The process therefore terminates at an optimum containing all G and no Z . $\square$

At $k = 25$ the screening rule fixes 4 occupations as included and excludes 574 occupations as impossible members without changing the coherent optimum.

V. EXACT INDEPENDENT-BOX REGRET OF A FIXED LIST

For a fixed set $X = \{i : x_i = 1\}$, $|X| = k$, define absolute box regret

$$R_{\square}^{\text{abs}}(X) = \max_{p \in \mathcal{P}_{\square}} \left[V_k(p) - \sum_{i \in X} p_i \right]. \quad (14)$$

The maximizing corner is

$$v_i(X) = \begin{cases} \ell_i, & i \in X, \\ u_i, & i \notin X, \end{cases} \quad (15)$$

so the exact closed form is

$$R_{\square}^{\text{abs}}(X) = V_k(v(X)) - \sum_{i \in X} \ell_i. \quad (16)$$

Writing the expression with V_k avoids ambiguity over the identity of a tied Top- k corner set.

VI. NESTED-SUPPORT CONTRACTION AND EXACT BOUNDARIES

Possibilistic and interval-weight approaches can parameterize uncertain decision inputs [25], [26]. Possibility theory is used only as optional parameterization language: the empirical object is a finite priority support plus a declared point core, not a calibrated possibility distribution. Retained support width $g \in [0, 1]$ defines

$$L_i(g) = c_i - g(c_i - \ell_i), \quad U_i(g) = c_i + g(u_i - c_i). \quad (17)$$

By (5), this is a nested family with $g = 1$ equal to the full box and $g = 0$ equal to the point core.

Rank vectors are piecewise constant. We denote each reported structural transition boundary by g^* . A best-rank comparison changes only at $L_j(g) = U_i(g)$,

$$g_{ji}^{(B)} = \frac{c_j - c_i}{(c_j - \ell_j) + (u_i - c_i)}, \quad (18)$$

and a worst-rank comparison only at $U_j(g) = L_i(g)$,

$$g_{ji}^{(W)} = \frac{c_i - c_j}{(u_j - c_j) + (c_i - \ell_i)}, \quad (19)$$

when the denominator is positive and the ratio lies in $[0, 1]$. Thus all events follow from a finite exact breakpoint set rather than Monte Carlo sampling.

TABLE II
REPORTED TOP-25 STRUCTURAL CONTRACTION BOUNDARIES.

Event	g^*	$1 - g^*$	Interpretation
Guaranteed $\rightarrow$ 14	.574320	.425680	coherent intersection cardinality; 32 shared point list
Possible $\rightarrow$ 37	.434651	.565349	
Full 25/25	.078756	.921244	

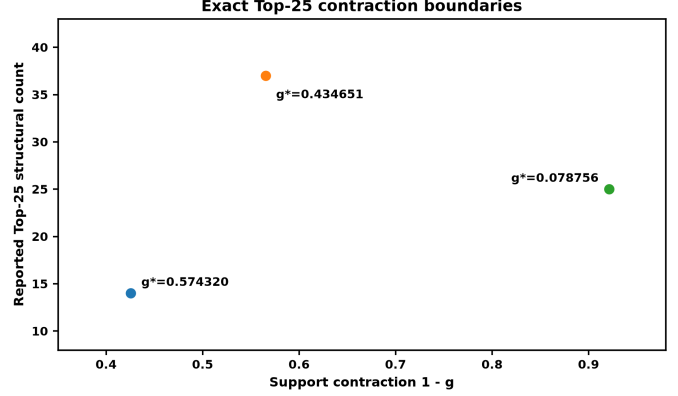


Fig. 2. Three reported exact Top-25 contraction boundaries. Points mark structural transition values; post-contraction membership follows the stated tie convention.

The endpoint convention matters. Since best rank uses $L_j > U_i$ while worst rank uses $U_j \geq L_i$, g^* is a *transition boundary*. When contraction proceeds from $g = 1$ toward zero, the post-contraction membership state applies for $g < g^*$; a newly strict best-rank comparison is not asserted at equality. Table II therefore reports boundaries rather than “first-at-equality” events.

For any strictly decreasing homeomorphism $\phi : [0, 1] \rightarrow [0, 1]$ with $\phi(0) = 1$ and $\phi(1) = 0$, setting $g = \phi(\alpha)$ traverses the same nested supports in the same order. Examples $\phi(\alpha) = 1 - \alpha$ and $\phi(\alpha) = (1 - \alpha)^2$ change numerical α labels but not the structural event sequence.

VII. APPLICATION RESULTS

Table III compares the three primary fixed-list rules and one transparent consensus benchmark. The relative-minimax list improves the minimum coherent-scenario capture to 96.52%, from 93.70% for the baseline composite and 88.35% for scale only. Its exact box absolute regret is 0.01945, slightly larger than the baseline’s 0.01810, so no list is unconditionally “most robust” without naming the uncertainty structure and criterion.

As a descriptive rank-aggregation benchmark, let $T_t = \text{Top}_{25}(p^t)$ denote the set of the 25 occupations with the largest priorities under scenario t , and define

$$f_i = \sum_{t=1}^{18} \mathbf{1}\{i \in T_t\}. \quad (20)$$

The Consensus-Frequency list contains the 25 occupations with the largest f_i . The 25th-ranked count is 9 and the

TABLE III

FIXED-LIST PERFORMANCE AT $k = 25$ OVER 18 PRIMITIVE SCENARIOS.

Rule	Mean cap.	Min. cap.	Max rel. regret	Box abs. regret
Scale only	93.02%	88.35%	11.65%	.02380
Baseline composite	96.14%	93.70%	6.30%	.01810
Consensus-Frequency	97.00%	95.89%	4.11%	.01952
Relative minimax	96.99%	96.52%	3.48%	.01945

26th-ranked count is 8, so no frequency tie occurs at the membership cutoff; ascending SOC is retained only as a fixed deterministic tie-break for equal counts away from the cutoff. Because the 18 scenarios are declared designs rather than probability-weighted draws, f_i is an occurrence count, not a selection probability. This simple benchmark is included only as a sanity check against broader rank/list-aggregation ideas [?], [?].

The Consensus-Frequency list overlaps the relative-minimax solution in 24 of 25 occupations and has a marginally higher mean capture (97.00% versus 96.99%), but its minimum capture is lower (95.89% versus 96.52%), corresponding to a larger maximum relative regret (4.11% versus 3.48%). Thus frequency aggregation closely reproduces the central list, while relative minimax provides stronger protection on the worst-case relative-capture criterion it explicitly optimizes; this is not a claim of universal dominance. Detailed scenario-by-scenario performance, the full 25-member list, and the one-item disagreement are retained in the reproducibility addendum; the one-item disagreement is also stated in the Supplement.

Criterion normalization also matters. Define coherent absolute regret

$$R_t^{\text{abs}}(x) = V_t^* - \sum_i p_i^t x_i, \quad (21)$$

and solve $\min_{x: \sum_i x_i = k, x_i \in \{0,1\}} \max_t R_t^{\text{abs}}(x)$. This alternative changes two of the 25 occupations: its minimum relative capture is 94.79% and maximum relative regret 5.21%, with 23/25 overlap with the relative-minimax list.

VIII. PRACTICAL INTERPRETATION AND LIMITS

The results show why occupational AI prioritization should not be treated as one fixed ranking. Some occupations remain priority candidates across all 18 admitted constructions, while 23 additional occupations enter or leave the Top-25 depending on measurement design. For analysts or institutions that require one fixed screening list, the relative-minimax solution is a defensible compromise because it maximizes the worst retained share of scenario-optimal priority mass across the declared designs. The 24/25 overlap with the simple Consensus-Frequency benchmark indicates close agreement between the two lists, while the lower worst-case regret identifies the specific protection provided by the minimax criterion.

The uncertainty family is finite and reflects admitted measurement choices rather than a probability law. Stochastic acceptability analysis and budgeted-uncertainty

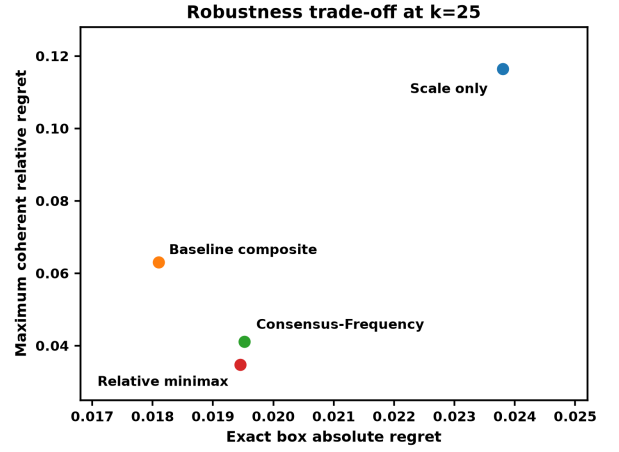


Fig. 3. Robustness trade-off at $k = 25$. Lower values are preferred on both axes. Consensus-Frequency lies close to relative minimax but has higher coherent worst-case regret; the baseline composite retains the smallest independent-box regret.

optimization address different decision questions [27], [28]. The present list is intended for screening and further assessment. It does not estimate AI-caused displacement, worker movement, training eligibility, transition capacity, or program success. Employment is a scale variable rather than a welfare weight.

IX. CONCLUSION

Across 661 U.S. occupations, AI-exposure disagreement materially affects who enters a Top-25 priority list. Yet the disagreement is structured: 14 occupations are stable across all coherent designs and 37 form the coherent candidate envelope. Exact interval ranks distinguish guaranteed from merely possible membership, while coherent minimax supplies one fixed list when a decision must be made. Relative minimax raises worst-scenario capture from 93.70% to 96.52% compared with the baseline composite and retains 24/25 occupations from the Consensus-Frequency benchmark, without claiming universal robustness. Exact breakpoint semantics prevent equality/tie conventions from being mistaken for new membership at a boundary. The contribution is therefore an applied occupational-screening result supported by exact optimization certificates, rather than an optimization method presented with a workforce example.

Data and Code Availability— The data, code, scenario matrices, selected lists, and reproducibility materials supporting this study are publicly available in the project [GitHub](#) ↗.

REFERENCES

- [1] M. Yin, H. Vu, and C. Persico, “How (un)Stable Are LLM Occupational Exposure Scores? Evidence from Multi-Model Replication,” NBER Working Paper 35110, 2026, doi: 10.3386/w35110.
- [2] E. W. Felten, M. Raj, and R. Seamans, “Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses,” *Strategic Management Journal*, vol. 42, no. 12, pp. 2195–2217, 2021, doi: 10.1002/smj.3286.

- [3] K. Tomlinson, S. Jaffe, W. Wang, S. Counts, and S. Suri, "Working with AI: Measuring the Applicability of Generative AI to Occupations," arXiv:2507.07935, 2025, doi: 10.48550/arXiv.2507.07935.
- [4] U.S. Bureau of Labor Statistics, *Occupational Employment and Wage Statistics, May 2025 National Estimates*, 2026.
- [5] U.S. Department of Labor, Employment and Training Administration, *O*NET 30.3 Database*, May 2026 release.
- [6] OECD and Joint Research Centre, *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing, 2008, doi: 10.1787/9789264043466-en.
- [7] C. F. Manski, *Partial Identification of Probability Distributions*. New York, NY, USA: Springer, 2003.
- [8] S. Greco, V. Mousseau, and R. Słowiński, "Ordinal regression revisited: Multiple criteria ranking using a set of additive value functions," *European Journal of Operational Research*, vol. 191, no. 2, pp. 416–436, 2008.
- [9] S. Steegen, F. Tuerlinckx, A. Gelman, and W. Vanpaemel, "Increasing transparency through a multiverse analysis," *Perspectives on Psychological Science*, vol. 11, no. 5, pp. 702–712, 2016.
- [10] U. Simonsohn, J. P. Simmons, and L. D. Nelson, "Specification curve analysis," *Nature Human Behaviour*, vol. 4, pp. 1208–1214, 2020.
- [11] M. A. Soliman and I. F. Ilyas, "Ranking with uncertain scores," in *Proc. IEEE ICDE*, 2009, pp. 317–328.
- [12] K. Mouratidis, J. Zhang, and H. Pang, "Maximum rank query," *Proceedings of the VLDB Endowment*, vol. 8, no. 12, pp. 1554–1565, 2015.
- [13] K. Mouratidis and B. Tang, "Exact processing of uncertain top-k queries in multi-criteria settings," *Proceedings of the VLDB Endowment*, vol. 11, no. 8, pp. 866–879, 2018.
- [14] D. Bertsimas and M. Sim, "Robust discrete optimization and network flows," *Mathematical Programming*, vol. 98, pp. 49–71, 2003, doi: 10.1007/s10107-003-0396-4.
- [15] D. Bertsimas and M. Sim, "The price of robustness," *Operations Research*, vol. 52, no. 1, pp. 35–53, 2004, doi: 10.1287/opre.1030.0065.
- [16] D. Bertsimas, D. B. Brown, and C. Caramanis, "Theory and applications of robust optimization," *SIAM Review*, vol. 53, no. 3, pp. 464–501, 2011, doi: 10.1137/080734510.
- [17] H. Aissi, C. Bazgan, and D. Vanderpooten, "Min-max and min-max regret versions of combinatorial optimization problems: A survey," *European Journal of Operational Research*, vol. 197, no. 2, pp. 427–438, 2009.
- [18] J. Ide and A. Schöbel, "Robustness for uncertain multi-objective optimization," *OR Spectrum*, vol. 38, pp. 235–271, 2016.
- [19] B. Roy, "Robustness in operational research and decision aiding: A multi-faceted issue," *European Journal of Operational Research*, vol. 200, no. 3, pp. 629–638, 2010.
- [20] I. Averbakh, "Computing and minimizing the relative regret in combinatorial optimization with interval data," *Discrete Optimization*, vol. 2, no. 4, pp. 273–287, 2005.
- [21] P. Kouvelis and G. Yu, *Robust Discrete Optimization and Its Applications*. Boston, MA, USA: Kluwer, 1997.
- [22] L. J. Savage, "The theory of statistical decision," *Journal of the American Statistical Association*, vol. 46, no. 253, pp. 55–67, 1951.
- [23] E. Conde, "An improved algorithm for selecting p items with uncertain returns," *Mathematical Programming*, vol. 100, no. 2, pp. 345–353, 2004.
- [24] A. Kasperski, *Discrete Optimization with Interval Data: Minmax Regret and Fuzzy Approach*. Berlin, Germany: Springer, 2008.
- [25] A. Kasperski and P. Zieliński, "Minmax regret approach and optimality evaluation in combinatorial optimization problems with interval and fuzzy weights," *European Journal of Operational Research*, vol. 200, no. 3, pp. 680–687, 2010, doi: 10.1016/j.ejor.2009.01.044.
- [26] D. Dubois and H. Prade, "Practical methods for constructing possibility distributions," *International Journal of Intelligent Systems*, vol. 31, no. 3, pp. 215–239, 2016, doi: 10.1002/int.21782.
- [27] R. Lahdelma, J. Hokkanen, and P. Salminen, "SMAA—stochastic multiobjective acceptability analysis," *European Journal of Operational Research*, vol. 106, no. 1, pp. 137–143, 1998, doi: 10.1016/S0377-2217(97)00163-X.
- [28] M. Goerigk and M. Khosravi, "Robust combinatorial optimization problems under budgeted interdiction uncertainty," *OR Spectrum*, vol. 47, pp. 255–285, 2025, doi: 10.1007/s00291-024-00772-0.

Supplementary Material

When AI Exposure Rankings Disagree: Robust Top-25 Selection Across 661 U.S. Occupations

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S1. Input and scenario register

The decision layer uses three frozen normalized exposure channels: AI Occupational Exposure (AIOE) [1], Microsoft user-goal, and Microsoft AI-action/nonphysical [2]. Employment scale is May 2025 OEWS employment share [3], and the five occupation-level skill descriptors are O*NET 30.3 Importance values [4]. The construction follows transparent composite-indicator practice [5].

Let $a_i^{(r)} \in [0, 1]$ denote frozen normalized exposure for channel r and $z_{im} \in [0, 1]$ the frozen normalized O*NET value for skill m . The five skills are Critical Thinking, Active Learning, Learning Strategies, Active Listening, and Speaking. For construction w with retained set M_w ,

$$h_i^{(w)} = \frac{1}{|M_w|} \sum_{m \in M_w} z_{im}, \quad q_i^{(r,w)} = a_i^{(r)} (1 - h_i^{(w)}), \quad p_i^t = s_i q_i^{(r,w)}.$$

The factor $1 - h_i^{(w)}$ is a declared screening transformation: a larger retained-skill aggregate lowers the modifier within this construction, but this does not establish a causal protective effect of skills. The equal-five construction uses all five skills; each of the five leave-one-out constructions removes exactly one skill. Thus the 18 primitive priority vectors are the Cartesian product of three exposure channels and six constructions. The upstream normalized values are treated as frozen inputs to this decision-layer paper and are not re-estimated here. The priority box is $\ell_i = \min_t p_i^t$, $u_i = \max_t p_i^t$; the point core is $c_i = p_i^{\text{AIOE, equal-five}}$, hence $\ell_i \leq c_i \leq u_i$.

Table 1: Primitive scenario construction.

Axis	Admitted levels
Exposure channel	AIOE; Microsoft user-goal; Microsoft AI-action/nonphysical
Skill construction	equal-five; leave Critical Thinking out; leave Active Learning out; leave Learning Strategies out; leave Active Listening out; leave Speaking out

S2. Membership counts and the 14 stable occupations

Table 2: Membership counts by selection size.

k	Scenario intersection	Scenario union	Box guaranteed	Box possible
10	7	16	2	43
25	14	37	4	87
50	32	73	14	159
100	68	135	27	244

At $k = 25$, the 14 occupations present in every coherent Top-25 are:

1. General and Operations Managers (11-1021);
2. Accountants and Auditors (13-2011);
3. Registered Nurses (29-1141);
4. Waiters and Waitresses (35-3031);
5. Janitors and Cleaners, Except Maids and Housekeeping Cleaners (37-2011);
6. Cashiers (41-2011);
7. Retail Salespersons (41-2031);
8. Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products (41-4012);

9. Bookkeeping, Accounting, and Auditing Clerks (43-3031);
10. Receptionists and Information Clerks (43-4171);
11. Secretaries and Administrative Assistants, Except Legal, Medical, and Executive (43-6014);
12. Office Clerks, General (43-9061);
13. Heavy and Tractor-Trailer Truck Drivers (53-3032);
14. Stockers and Order Fillers (53-7065).

These are stable screening priorities within the declared 18-specification family; the result is not a displacement or program recommendation.

S3. Exact rank proof and tie convention

Best rank is existential:

$$r_i^{\min} = 1 + \sum_{j \neq i} \mathbf{1}\{\ell_j > u_i\}.$$

Worst rank is pessimistic:

$$r_i^{\max} = 1 + \sum_{j \neq i} \mathbf{1}\{u_j \geq \ell_i\}.$$

Here $\mathbf{1}\{\cdot\}$ denotes the indicator function. Possible membership is $r_i^{\min} \leq k$ and guaranteed membership is $r_i^{\max} \leq k$. The strict/non-strict distinction controls boundary inclusion in the contraction analysis.

S4. Safe-screening proof detail

Let $G = \{i : r_i^{\max} \leq k\}$ and $Z = \{i : r_i^{\min} > k\}$. Order the items in G by nonincreasing lower bound and insert them in that order. If the next $a \in G$ is absent from a size- k optimum, at most $k - 1$ competitors satisfy $u_j \geq \ell_a$, so the list contains an h with $u_h < \ell_a$. Any guaranteed item a' inserted earlier has $\ell_{a'} \geq \ell_a$ and hence $u_{a'} \geq \ell_{a'} \geq \ell_a$, so it cannot be chosen as h . Replacing h by a therefore preserves all earlier guaranteed insertions and weakly increases every scenario-wise selected mass.

After all G items have been inserted, suppose a selected $i \in Z$ remains. At least k competitors satisfy $\ell_j > u_i$, so at least one such j is omitted. Replacing i by that j strictly increases every scenario-wise selected mass. Because the family of size- k lists is finite, strict exchanges cannot cycle. If a Z item remained, another strict exchange would exist; therefore the process terminates at an optimum containing all G and no Z . This establishes the existence claim for every objective coordinatewise nonincreasing in scenario-wise selected mass.

S5. Fixed-list verification, consensus benchmark, and criterion sensitivity

Table 3: Fixed-list comparison at $k = 25$ over 18 primitive scenarios.

Rule	Mean capture	Minimum capture	Max. rel. regret	Box regret
Scale only	93.02%	88.35%	11.65%	0.02380
Baseline composite	96.14%	93.70%	6.30%	0.01810
Consensus-Frequency	97.00%	95.89%	4.11%	0.01952
Relative minimax	96.99%	96.52%	3.48%	0.01945

For the Consensus-Frequency benchmark, let $T_t = \text{Top}_{25}(p^t)$ denote the 25 occupations with the largest priorities under scenario t . Then $f_i = \sum_{t=1}^{18} \mathbf{1}\{i \in T_t\}$ counts appearances across the 18 scenario-specific Top-25 lists. The 25 largest counts are selected; the cutoff is strict (9 appearances for the 25th occupation versus 8 for the 26th), so no frequency tie occurs at the membership cutoff. Ascending SOC is retained only as a fixed deterministic tie-break for equal counts away from the cutoff. The list overlaps relative minimax in 24/25 occupations: consensus includes Hosts and Hostesses, Restaurant, Lounge, and Coffee Shop (35-9031), whereas minimax includes Amusement and Recreation Attendants (39-3091). Exact scenario-by-scenario capture/regret are retained in the reproducibility addendum. Counts are design occurrences, not probabilities.

Consensus-Frequency Top-25 SOC membership. 11-1021, 13-2011, 29-1141, 35-3031, 37-2011, 41-2011, 41-2031, 41-4012, 43-3031, 43-4171, 43-6014, 43-9061, 53-3032, 53-7065, 13-1111, 43-1011, 13-1071, 15-1232, 33-9032, 35-2014, 35-9031, 41-1011, 43-5071, 43-6013, and 49-9071. The complete labeled list is supplied in `consensus_frequency_selection.csv`. The relative-minimax list contains the same 24 shared occupations, replacing Hosts and Hostesses, Restaurant, Lounge, and Coffee Shop (35-9031) with Amusement and Recreation Attendants (39-3091).

The relative-minimax objective is $\max_t R_t(x)$ with $R_t(x) = 1 - \sum_i p_i^t x_i / V_t^*$ and $V_t^* > 0$ for every admitted scenario. The absolute-minimax sensitivity instead defines

$$R_t^{\text{abs}}(x) = V_t^* - \sum_i p_i^t x_i, \quad x: \min_{x_i=k, x_i \in \{0,1\}} \max_t R_t^{\text{abs}}(x).$$

Its list overlaps the relative-minimax list in 23/25 occupations, with minimum relative capture 94.79% and maximum relative regret 5.21%.

S6. Nested-support sensitivity and exact boundary semantics

For retained width $g \in [0, 1]$,

$$[L_i(g), U_i(g)] = [c_i - g(c_i - \ell_i), c_i + g(u_i - c_i)].$$

We denote each reported structural transition boundary by g^* . These values are transition boundaries. Under the stated tie

Table 4: Structural Top-25 transition boundaries.

Event	g^*	$1 - g^*$	Identity check
Guaranteed $\rightarrow$ 14	0.5743200867	0.4256799133	coherent-intersection set after crossing
Possible $\rightarrow$ 37	0.4346510223	0.5653489777	cardinality only; 32 shared
Full 25/25	0.0787564209	0.9212435791	point list

convention, the post-contraction state applies for $g < g^*$ when contraction proceeds downward from $g = 1$; equality is evaluated by the original strict $>$ and non-strict $\geq$ rules rather than by rounded prose labels.

At the 37-member boundary, the five possible-set-only occupations are Computer and Information Systems Managers (11-3021), Medical Assistants (31-9092), Billing and Posting Clerks (43-3021), First-Line Supervisors of Construction Trades and Extraction Workers (47-1011), and Inspectors, Testers, Sorters, Samplers, and Weighers (51-9061). The five coherent-union-only occupations are Hosts and Hostesses, Restaurant, Lounge, and Coffee Shop (35-9031), Amusement and Recreation Attendants (39-3091), Counter and Rental Clerks (41-2021), Production, Planning, and Expediting Clerks (43-5061), and First-Line Supervisors of Production and Operating Workers (51-1011).

For a strictly decreasing homeomorphism $\phi : [0, 1] \rightarrow [0, 1]$ with $\phi(0) = 1$, $\phi(1) = 0$, set $g = \phi(\alpha)$. Linear $\phi(\alpha) = 1 - \alpha$ and quadratic $\phi(\alpha) = (1 - \alpha)^2$ relabel the same support family and cannot constitute independent robustness tests.

S7. Claim-admissibility matrix

Table 5: Permitted and unsupported interpretations.

Output	Permitted	Unsupported
Box guaranteed/possible	Exact membership under independent priority intervals	Probability of selection
Coherent intersection/union	Membership over 18 admitted designs	Universal robustness
Relative-minimax list	Best worst relative capture over declared scenarios	Optimal workforce policy
Consensus-frequency list	Occurrence-based aggregation over 18 admitted Top-25 lists	Probability of selection or worst-case optimum
Box regret	Exact absolute regret under independent priority intervals	Dominance under coherent designs
Boundary g^*	Width at which rank-comparison status can change	Confidence or calibrated possibility

References

- [1] E. W. Felten, M. Raj, and R. Seamans, “Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses,” *Strategic Management Journal*, vol. 42, no. 12, pp. 2195–2217, 2021, doi: 10.1002/smj.3286.
- [2] K. Tomlinson, S. Jaffe, W. Wang, S. Counts, and S. Suri, “Working with AI: Measuring the Applicability of Generative AI to Occupations,” arXiv:2507.07935, 2025, doi: 10.48550/arXiv.2507.07935.
- [3] U.S. Bureau of Labor Statistics, *Occupational Employment and Wage Statistics, May 2025 National Estimates*, 2026.
- [4] U.S. Department of Labor, Employment and Training Administration, *O*NET 30.3 Database*, May 2026 release.
- [5] OECD and Joint Research Centre, *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing, 2008.